EX.No.-2.2 CREATE A 3D SOLID MODEL FROM ISOMETRIC DRAWING

AIM:

To create a 3D model from the given isometric drawing using the Fusion 360 tool.

REQUIREMENTS:

1. Autodesk Fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommended: 1 GB or more)
4. 2.5 Mbps or faster download speed; 500 Kbps or faster upload speed

PROCEDURE:

Step 1: Start the Sketch Command

- Open Autodesk Fusion 360 and select **Create Sketch**.

Step 2: Select the Sketch Plane

- Select the **TOP (XY)** Plane.

Step 3: Create the Base Sketch

1. Navigate to **Sketch > Rectangle > Center Rectangle**.
2. Hover over the origin (center) until the cursor snaps to the location.
3. Click once to place the center of the rectangle.
4. Press the **Tab** button to switch between input fields, then enter **56 mm** for width and **18 mm** for height.
5. Select **Sketch > Arc > Center Point Arc**, and place arcs at both ends of the sketch by snapping to the points.
6. Enter **R 18 mm** for the arc radius.
7. Select **Sketch > Circle > Center Point Circle**, place circles on both sides, and set the diameter to **dia 16 mm**.
8. Click **Finish Sketch**.

Step 4: Extrude the Base Sketch in the XY Plane

1. Select **Extrude**.
2. Select the base sketch profile.
3. Enter an extrusion value of **8 mm**.
4. Press **Enter** to accept and complete the extrusion.

Step 5: Start a New Sketch

- Select **Sketch > Create Sketch**.

Step 6: Select the Top Surface Plane

- Click on the **top surface** of the existing 3D block to use as your new sketching plane.

Step 7: Create a Rectangle Profile

1. Select **Sketch > Rectangle > Center Rectangle**.
2. Snap to the origin (center) of the sketch.
3. Enter dimensions of **40 mm** (width) and **10 mm** (height) using the **Tab** key.
4. Click to place the rectangle, then click **Finish Sketch**.

Step 8: Extrude the Rectangle

1. Select **Extrude**.
2. Select the rectangle sketch profile.
3. Enter an extrusion distance of **32 mm**.
4. Press **Enter**.

Step 9: Create a Triangle Profile

1. Select **Sketch > Triangle > Center Triangle**.
2. Snap to the center origin.
3. Enter dimensions of **32 mm** (width) and **12 mm** (height).
4. Click to place the triangle, then click **Finish Sketch**.

Step 10: Extrude the Triangle

1. Select **Extrude**.

2. Select the triangle sketch.
3. Set the extrusion to **6 mm** using the **Symmetric / Both Directions** option.
4. Press **Enter**.

Step 11: Create a Polygon (Pentagon/Triangle base configuration)

1. Select **Sketch > Polygon > Circumscribed Polygon**.
2. Snap to the origin.
3. Enter **40 mm** and **30 mm** for the width and height, and select **3 sides**.
4. Click to place the polygon, then click **Finish Sketch**.

Step 12: Extrude the Polygon

1. Select **Extrude**.
2. Select the polygon profile.
3. Enter an extrusion value of **10 mm**.
4. Press **Enter**.

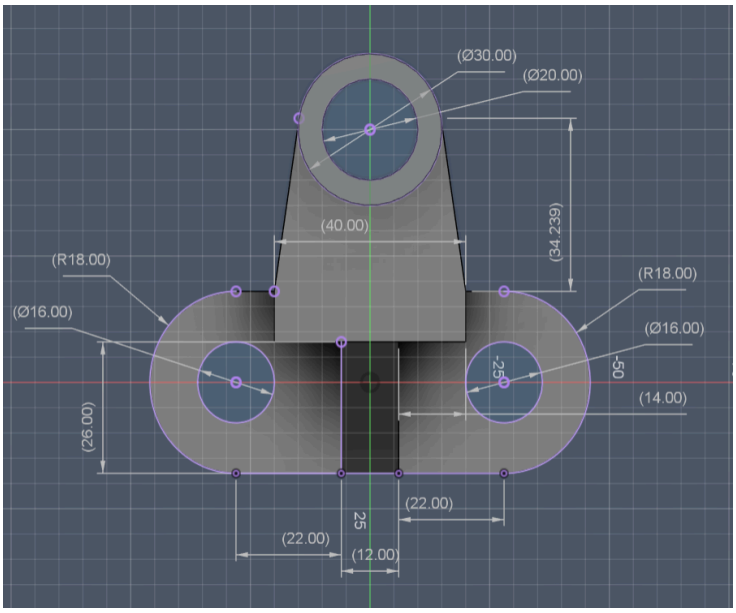
Step 13: Create a Circle Feature

1. Select **Sketch > Circle > Center Circle**.
2. Snap to the end side of the polygon shape.
3. Enter an inner diameter of **20 mm** and an outer diameter of **dia 30 mm**.
4. Click to place the circle, then click **Finish Sketch**.

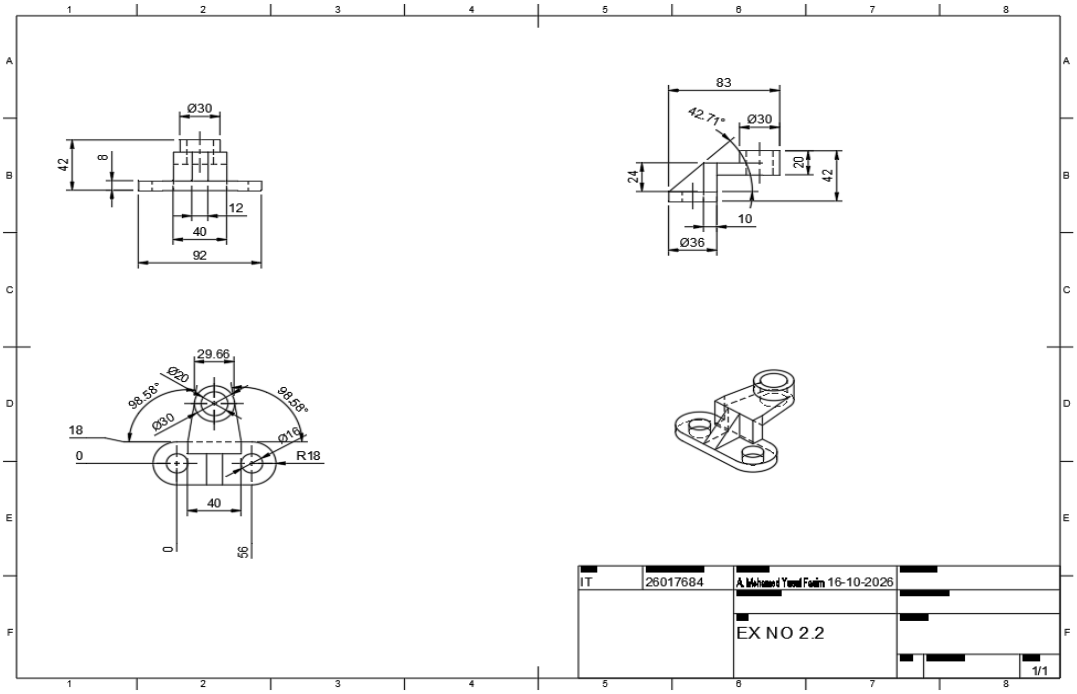
Step 14: Extrude the Circle

1. Select **Extrude**.
2. Select the circle sketch profile.
3. Enter an extrusion value of **10 mm** in **Both Sides** direction.
4. Press **Enter**.

Input:



Output:



RESULT:

Thus, the required 3D solid model has been successfully created from the given isometric drawing using Autodesk Fusion 360.